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MAJOR ARTICLE

Clinical Manifestations of Nipah Virus–Infected Patients Who Presented to the Emergency Department During an Outbreak in Kerala State in India, May 2018

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Background. An outbreak of Nipah virus (NiV) disease occurred in the Kozhikode district of Kerala State in India in May 2018. Several cases were treated at the emergency medicine department (ED) of the Government Medical College, Kozhikode (GMCK). The clinical manifestations and outcome of these cases are described.

Methods. The study included 12 cases treated in the ED of GMCK. Detailed clinical examination, laboratory investigations, and molecular testing for etiological diagnosis were performed.

Results. The median age of the patients was 30 years and the male to female ratio was 1.4:1.0. All the cases except the index case contracted the infection from hospitals. The median incubation period was 10 days, and the case fatality ratio was 83.3%. Ten (83.3%) patients had encephalitis and 9 out of 11 patients whose chest X-rays were obtained had bilateral infiltrates. Three patients had bradycardia and intractable hypotension requiring inotropes. Encephalitis, acute respiratory distress syndrome, and myocarditis were the clinical prototypes, but there were large overlaps between these. Ribavirin therapy was given to a subset of the patients. Although there was a 20% reduction in NiV encephalitis cases treated with the drug, the difference was not statistically significant. The outbreak ended soon after the introduction of total isolation of patients and barrier nursing.

Conclusion. The outbreak of NiV disease in Kozhikode in May 2018 presented as encephalitis, acute respiratory distress and myocarditis or combinations of these. The CFR was high. Ribavirin therapy was tried but no evidence for its benefit could be obtained.

Keywords. Nipah virus; encephalitis; Nipah epidemic; Nipah clinical manifestations Kozhikode; Kerala-India.

Nipah virus (NiV) infection is a newly emerging viral zoonosis that has the potential to cause severe disease in both animals and humans [1]. NiV, a member of the *Henipavirus* genus of the Paramyxoviridae family [2], is a highly infectious pathogen [3]. The natural host of the virus is the fruit bat of the *Pteropus* genus belonging to the Pteropodidae family [4]. NiV was implicated as the causative agent of outbreaks in humans and pigs in Malaysia in 1998 [5, 6], where pigs acted as intermediate amplifying hosts [7].

The disease re-emerged in Bangladesh in 2001 and again in 2003 [8]. An epidemiological link to the consumption of raw date palm juice was identified during an outbreak in 2005 [9]. Twenty-three introductions of NiV from animal carriers to humans resulting in 10 outbreaks and 122 cases were identified in Bangladesh between 2001 and 2007 [10]. Drinking raw date palm juice and contact with patients with NiV disease, including caring for such patients, were the most common risk factors [11].

In India, NiV infection was first detected when clinical specimens of patients affected during an outbreak of febrile illness with altered sensorium, which occurred in Siliguri, West Bengal, in 2001, were tested retrospectively [12, 13]. Forty-five (75%) of the 60 cases were hospital workers who cared for NiV cases or who had exposure to a patient with NiV disease in a hospital. The outbreak ended 5 days after the introduction of barrier nursing in the hospitals [12]. Another cluster of 5 cases of NiV encephalitis occurred in a village on the Bangladesh border in 2007 [14].

On 5 May 2018, a young male patient died of encephalitis of unknown etiology at Government Medical College, Kozhikode (GMCK), in the state of Kerala, India. Within 2 weeks of his death, 3 of his family members developed similar illness and all of them died. Blood samples of these patients tested positive for NiV RNA at the Manipal Institute of Virology (MIV), formerly the Manipal Centre for Virus Research, and this was confirmed by the Indian Council of Medical Research (ICMR)-National Institute of Virology, Pune, India. The NiV etiology of the disease was known just before the death of the second case [15].

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Because of the fear of nosocomial transmission of the infection, the referral services at GMCK were restricted to emergency cases. An isolation ward for patients with NiV infection with barrier nursing and other precautionary measures was set up in the emergency medicine department (ED), where all further cases were treated. Here we present the clinical manifestations and course of these cases.

METHODS

A total of 13 patients with NiV infection were treated in GMCK, out of which 12 patients were treated at the ED and were included in this study. One of the 13 patients was admitted with poisoning to GMCK and became infected while in the intensive care unit and hence not included. NiV infection in all of these patients, except for the index case, was proven by real-time reverse-transcriptase polymerase chain reaction (PCR) on either or a combination of serum/plasma, urine, cerebrospinal fluid (CSF), throat swab, and endotracheal aspirate targeting the nucleocapsid gene of NiV at MIV and National Institute of Virology, Pune, as described by Arunkumar et al [15].

All patients were subjected to detailed clinical examination and laboratory and radiological assessment. Complete blood count, blood glucose, renal and liver function tests, arterial blood gas (ABG) analysis, and serum troponin I level determinations were done in all patients. Chest X-ray and bedside screening echocardiography were performed in selected patients. Troponin I levels were assessed quantitatively by chemiluminescence immunoassay, and a value of 0.04 ng/mL or greater was considered elevated and indicative of myocarditis. Partial Pressure of Oxygen (PO_2) less than 88 mm Hg in ABG was considered hypoxemia. Bedside screening echocardiography was performed with a Sonosite m-turbo ultrasound machine (Fujifilm SonoSite, Inc) using a phased-array probe with a frequency of 1–5 MHz.

Patient Management

Fever triaging was done at the ED. Patient management focused on intensive supportive care with augmented infection-control practices. Encephalitis was managed with treatment for raised intracranial pressure and seizure. Noninvasive positive-pressure ventilation or mechanical ventilation was given when acute respiratory distress syndrome (ARDS) was present. All patients were treated with antibiotics and antivirals (acyclovir/oseltamivir) to cover the usual bacterial, rickettsial, and viral etiologies of encephalitis in the region. Ribavirin therapy as an empirical anti-NiV strategy was introduced subsequently in the dose of 2 g on admission followed by 1 g every 6 hours for 4 days and 500 mg every 6 hours for 6 days orally.

Statistical Analysis

Continuous variables were summarized with medians and interquartile ranges (IQRs). Binomial variables were summarized with proportions and mid-p confidence intervals CIs). The statistical significance of difference in proportions was tested using Fisher's exact test. OpenEpi (https://www.openepi.com/Proportion/Proportion.htm) was used to calculate the CIs of proportions and to do Fisher's exact test. SPSS version 25.0 was used to calculate the medians and IQRs.

RESULTS

Patients and Diagnosis

The case series reported here includes the index case and all of the patients with laboratory-confirmed NiV infection treated at the ED of GMCK during the outbreak in May 2018. The age of the patients was in the range of 19–49 years, with a median of 30 years (IQR, 25.25–47.25 years). Seven of the 12 patients were males; thus, the male to female ratio was 1.4:1.0. Real-time reverse-transcriptase PCR was positive in at least 1 of the samples, such as serum/plasma (10 tested and 8 positive), CSF (6 tested and 3 positive), urine (9 tested and 7 positive), throat swab (7 tested and 5 positive) and endotracheal aspirate (3 tested and all positive).

Source of Infection

All of the cases except for the index case contracted the infection from other confirmed cases of NiV infection through exposure within hospitals. In 9 cases, the source was the index case. All of the exposures resulting in infection occurred before the NiV etiology of the outbreak became known. Transmission in all of the cases, except for the index case, was nosocomial [15]. Two (18.2%) of the 11 cases who acquired the infection from the index case were hospital workers.

Clinical Course and Outcome

Table 1 summarizes demographic, clinical, and laboratory parameters of cases of NiV disease on the day of presentation to the ED. The estimated incubation period ranged between 8 and 15 days, with a median of 10 days (IQR, 8.0–12.0 days). The median duration of hospital stay before death in the fatal cases was 2 days, with a maximum of 4 days. Ten of the 12 patients died; thus, the case fatality ratio (CFR) was 83.3% (95% CI, 54.9–97.1%). Serial testing of samples by reverse-transcriptase PCR was done only in the 2 cases who survived. In the first case, a 19-year-old female, throat swab and urine tested positive on the day of admission to the ED, which was the sixth day after disease onset. The blood sample tested positive on day 9, and urine was negative. Blood tests were additionally performed on the 12th, 14th, and 24th day after disease onset and all tests were negative. A throat swab was repeated on the 24th day and was also negative. In the second case, a 27-year-old male, the throat swab tested positive on the second day. On the fifth day, blood and urine samples were negative; on the seventh day, the blood sample was positive, and on the ninth day blood and urine samples were positive. On the 10th, 11th, and 40th days, both blood and urine samples were negative

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Table 1. Demographic, Clinical, and Laboratory Parameters of Nipah Virus Disease Cases on Day of Presentation to the Emergency Department, Government Medical College, Kozhikode, Kerala, India, May 2018

Parameter	No.	Min	Max	Median	IQR	
Duration of fever before admission	12	1	7	4	1	7
Incubation period (except index case)	11	8	15	10	8	12
Duration of illness in fatal cases	10	3	10	6	5	8
Duration of hospital stay in fatal cases	10	0	4	2	2	3
Age	12	19	49	30	25.25	47.25
Heart rate at admission per minute	12	90	150	115	103	126
Systolic BP, mm Hg	12	110	190	120	110	145
Diastolic BP, mm Hg	12	60	100	80	72.5	90
Saturation in room air, %	12	60	100	86	78.5	96
Platelet count, × 10^3/μL	11	74	299	132	76.8	190
Total leucocyte count, per μL	11	4300	9200	6700	5900	8800

Abbreviations: BP, blood pressure; IQR, interquartile range; Max, maximum; Min, minimum.

and the throat swab repeated on the 40th day was negative. Semen sample of a 27 years old male survivor tested positive on day 16, and 26 after illness onset, and tested negative from the 42nd day of illness [16].

Frequencies of Symptoms/Signs

The frequencies of symptoms, signs, and abnormal laboratory parameters are summarized in Table 2. Ten (83.3%; 95% CI, 54.9–97.1%) patients had altered sensorium indicative of encephalitis and 3 (25%; 95% CI, 6.8–54.1%) had seizures. Two patients, including 1 patient who eventually recovered, had only fever at presentation. One of the surviving patients had fever, cough, and cervical lymphadenopathy without any signs of nervous system involvement, while the other patient later progressed to ARDS and encephalitis and expired.

Table 2. Frequencies of Symptoms, Signs, and Abnormal Laboratory Parameters Among the 12 Cases of Nipah Virus Infection Treated at the Emergency Department, Government Medical College, Kozhikode, India, May 2018

Symptom/Sign/Laboratory Parameter	No.	Present	%
Fever	12	12	100.0
Vomiting	12	7	58.3
Headache	12	9	75.0
Cough	12	4	33.3
Breathlessness	12	5	41.7
Altered sensorium	12	10	83.3
Convulsions	12	3	25.0
Tachypnea	12	9	75.0
Crepitations	12	10	83.3
Hypoxia in ABG analysis	12	9	75.0
Infiltrates in chest X-ray	11	9	81.8
Ventilatory support	12	11	91.7
Platelet count <140 × 10^3 cells per microliter	11	6	54.5
Platelet count <100 × 10^3 cells per microliter	11	3	27.3
LV hypokinesia in echocardiogram	7	3	42.9
Developed bradycardia	12	3	25.0
Developed hypotension	12	4	33.3
Required inotropes	12	3	25.0

Abbreviations: ABG, arterial blood gas; LV, left ventricular.

Although cough was present in only 4 patients and breathlessness in 5, on chest auscultation, crepitations were audible in 10 (83.3%; 95% CI, 54.9–97.1%) and ABG analysis showed hypoxia in 9 patients (75%; 95% CI, 45.9–93.2%). Chest X-ray was obtained in 11 patients, and in 9 (81.8%; 95% CI, 51.7–96.8%), bilateral infiltrates were visible. All patients who had altered sensorium also had signs of respiratory system involvement. Eleven of the 12 patients needed ventilatory support.

Severe headache and vomiting were the other most frequent symptoms, which were present in 9 (75%; 95% CI, 45.9–93.2%) and 7 (58.3%; 30.2–82.8%) patients, respectively. Three (25%; 95% CI, 6.8–54.1%) patients developed bradycardia and intractable hypotension and needed administration of inotropes. Echocardiogram was performed in 7 patients, and 3 showed left ventricular (LV) hypokinesia. However, these patients did not develop hypotension or bradycardia. Echocardiogram was performed in only one of the patients who developed hypotension and in that patient there was no LV hypokinesia. Platelet count was performed in 11 patients and, among them, 6 had counts less than 140 × 10^3 cells per microliter and 3 had counts less than 100 × 10^3/μL. Table 1 summarizes the median values of clinical measurements and quantitative laboratory parameters. The notable abnormalities were tachycardia and thrombocytopenia. The median values of blood pressure, both systolic and diastolic, were within normal limits. Other manifestations observed were blanching rash, cervical lymph node enlargement, diarrhea, melena, hemoptysis, and elevated levels of alanine aminotransferase and aspartate aminotransferase, all seen in 1 patient each.

Cardiovascular Manifestations

Tachycardia was present in 10 patients (83.3%; 95% CI, 54.9–97.1%) and hypertension and ST segment and T wave (ST-T) changes in the initial electrocardiogram in 3 patients (25%; 95%

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CI, 6.8–54.1%). Bedside screening echocardiogram was suggestive of LV hypokinesia in 3 patients (25%; 95% CI, 6.8–54.1%) and elevated troponin I values were noted in 4 patients (33.3%; 95% CI, 11.6–62.3%).

Syndromic Presentations

Encephalitis, ARDS, and myocarditis were the clinical phenotypes observed in the patients. Encephalitis and ARDS were present in 10 patients each, and among these, 9 patients had both ARDS and encephalitis. Nine out of the 10 fatal cases had encephalitis and ARDS, while one had neither at admission but developed both encephalitis and ARDS during his hospital stay. Among the 2 nonfatal cases, one had both encephalitis and ARDS while the other did not have either. Thus, both encephalitis and ARDS were present in 90% of fatal cases but only in 50% of the nonfatal cases, although the number is too small to draw any definite inference. Myocarditis was present in 5 patients, of whom one had it along with encephalitis, one with ARDS and the remaining three had it along with encephalitis and ARDS. The survivors had a different spectrum of clinical presentation, with one having a combination of encephalitis, ARDS, and myocarditis and the other having only an upper respiratory illness.

Ribavirin Therapy

Ribavirin, a purine analog, was empirically tried in 6 patients, and among them, 2 survived, whereas all of the 6 patients who did not receive ribavirin succumbed to disease (Fisher's exact 1-tailed $P = .2273$). One of the surviving patients did not have encephalitis, and therefore the CFR observed in cases with NiV encephalitis treated with ribavirin in this case series was 80%, or the reduction in mortality was 20% (Fisher's exact 1-tailed $P = .4545$). However, in the absence of statistical significance for the difference in the CFRs, the effect of ribavirin therapy on the survival of patients with NiV remains unresolved.

DISCUSSION

The outbreak of NiV disease was the third reported in India. The previous outbreaks were in Siliguri (2001) and Nadia (2007) districts [12, 14], where the NiV etiology was identified after the outbreak ended [12, 14]. In the present outbreak, the etiology was confirmed in the second case [15]. However, the delay between the disease onset in the index case and detection of NiV etiology was 20 days.

All cases except for the index case acquired the infection from hospitals where they were exposed to other patients with NiV. Disease transmission ended soon after the protocols of total isolation of cases and barrier nursing were instituted.

Human-to-human transmission of NiV was considered to be rare in the Malaysian outbreak [17], although NiV could be isolated from throat swabs, nasal swabs, and urine of patients [18] and anti–NiV immunoglobulin (Ig) G antibodies were detected in a few hospital workers. However, these were considered to be false-positive results because neither IgM nor neutralizing antibodies could be detected in these individuals [19]. During the outbreak in Bangladesh in 2001, living with or caring for patients with NiV disease was found to be a significant risk factor for acquiring infection [8]. However, none of the healthcare workers who had cared for patients with NiV disease had anti-NiV antibodies and 40% of them had reported using barrier precautions. During the outbreak in Siliguri, 75% of the affected persons had a history of hospital exposure to a case with NiV disease, and among the 66 cases identified, at least 25 (37.9%) were hospital workers [12]. The outbreak ended 5 days after introducing barrier methods for nursing care.

The Bangladesh strain of NiV, which was similar to the strain isolated from Siliguri [14], was found to shed in significantly higher levels than the Malaysian strain in oral secretions of experimentally infected ferrets [20]. However, uninfected ferrets, when co-housed with those experimentally infected with either of the strains, did not contract the infection unless assisted by direct exposure to oronasal secretions of infected ferrets [21], suggesting that direct exposure to the secretions is a stronger determinant of transmissibility of infection. The strain of NiV that caused the outbreak in Kerala has been found to be closely related to the Bangladesh strain [22]. Primate studies have shown that the Bangladesh strain of NiV causes more severe pathological changes in the organs and higher mortality than the Malaysian strain [23].

The estimated incubation period was similar to that estimated during the outbreak in Malaysia [24]. The CFR observed during the present outbreak was 83.3%, while that observed during the first outbreak of NiV infection in Malaysia was 32% [25]. The Malaysian case series had a broad case definition and included both laboratory-confirmed and clinically and epidemiologically probable cases as well as cases with and without encephalitis. Besides the strain difference, this might also have contributed to the lower CFR during the Malaysian outbreak. Among the group of 52 patients of the Malaysian outbreak who had central nervous system involvement, 32 died [25], a CFR of 61.5%. The overall CFR during all the NiV epidemics reported in Bangladesh during the period 2001–2011 was higher (77%) [11] and comparable to the CFR of the present outbreak.

Isolation of NiV from CSF has been shown to be associated with high mortality in NiV encephalitis [26], which is thought to be due to high viral replication in the central nervous system. NiV RNA was detected in 1 or more clinical specimens in all the cases except for the index case in the present outbreak. In the first outbreak of NiV infection in Siliguri, India, where altered sensorium/encephalitis was an essential inclusion criterion, the CFR was about 74% [12].

In the present outbreak, ARDS was present in 9 (75%) cases on admission. Even though in the series of 94 cases in Malaysia described by Goh et al [25] all systems except for the nervous

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system were normal at admission, among 103 cases from the same outbreak treated at Seremban Hospital, chest X-ray abnormalities were observed in 24 (57.1%) of the 42 cases [24] and in 19 of them alveolar consolidation was present. Among 11 cases of NiV encephalitis that occurred among abattoir workers in Singapore, 8 had abnormal chest radiographs with focal or diffuse interstitial shadowing and 2 had atypical pneumonia [27]. In a series of 92 cases from 4 outbreaks that occurred during 2001–2004 in Bangladesh, 69% had breathing difficulty and all of the 5 patients whose chest X-rays were obtained had images consistent with ARDS [28]. During the 2001 outbreak in Siliguri, 51% of the patients had respiratory manifestations ranging from tachypnea to acute respiratory distress [12]. In our series, chest X-rays were performed in 11 patients and, among them, 9 showed infiltrates (81.8%). These studies indicate that, as seen during the present case series, respiratory system involvement is quite common in NiV infection and the frequency might be higher in infections with the Bangladesh strain of NiV.

Various combinations of encephalitis, ARDS, and myocarditis were observed in this outbreak, but a majority of the patients had encephalitis and ARDS. Antiviral drugs such as acyclovir and oseltamivir did not appear to alter the course of the disease. The purine analog ribavirin was used towards the later stages of the outbreak. The drug could have reduced the mortality rate by about 20%, although no statistical significance could be demonstrated. In an open-label trial during the outbreak in Malaysia, ribavirin was found to reduce the mortality rate of NiV encephalitis by 36% [29].

Eleven of the 12 cases described in this series were the result of nosocomial transmission of NiV from the first case. Nine of these 11 cases acquired the infection in a span of 2 days. The spillover of the virus from non human source happened only once. The delay between the disease onset in the index case and the recognition of the risk of nosocomial transmission was 20 days. All the exposure events had occurred within this period.

No treatment strategy appears to be effective in reducing the mortality in NiV encephalitis, although NiV infection without encephalitis may have a much lower mortality. Various antiviral strategies, including attachment competitors, fusion inhibitor peptides, and passive immunotherapy, are being explored [30], but none have been successful so far. Prevention of nosocomial transmission is the best strategy to limit the spread of NiV outbreaks.

In conclusion, this is the first comprehensive report on clinical manifestations of NiV diseases from India and there is a need for more clinical data to understand the disease process and to design more efficient clinical case-management strategies.

Supplementary Data

Supplementary materials are available at *Clinical Infectious Diseases* online. Consisting of data provided by the authors to benefit the reader, the posted materials are not copyedited and are the sole responsibility of the authors, so questions or comments should be addressed to the corresponding author.

Notes

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